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PROJECT INITIATION

README.md Template

BrainBytes AI Tutoring Platform

Project Overview

BrainBytes is an AI-powered tutoring platform designed to provide accessible academic assistance to Filipino students. This project implements the platform using modern DevOps practices and containerization. It provides intelligent tutoring across multiple subjects including Math, Science, History, English, and Filipino.

Team Members

- **Eldan Eunice Sinsuan** - Team Lead [eldan.sinsuan@mmdc.mcl.edu.ph]
- **Sarah Nicole Hular** - Frontend Developer [sarah.hular@mmdc.mcl.edu.ph]
- **Mara Julienne Rose Cervantes** - Frontend Developer [mara.cervantes@mmdc.mcl.edu.ph]
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- **Michelle Joi Quesada** - DevOps Engineer [michelle.quesada@mmdc.mcl.edu.ph]



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Project Goals

- Implement a containerized multi-container application with proper networking
- Integrate AI-powered tutoring using Hugging Face Inference API
- Create a database-driven application with MongoDB for storing user profiles, learning materials, and chat history
- Apply MVC architecture for clean and maintainable backend code
- Deploy using Docker Compose with frontend, backend, and database containers

Technology Stack

- **Frontend**: Next.js (React)
- **Backend**: Node.js with Express.js (MVC Architecture)
- **Database**: MongoDB
- **AI Integration**: Hugging Face Inference API with intelligent fallback
- **Containerization**: Docker & Docker Compose
- **Architecture**: Multi-container setup with 3 services

Architecture



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The application consists of three Docker containers:

- ****Frontend**** (Next.js) - User interface running on port 8080
- ****Backend**** (Express.js) - REST API running on port 3000
- ****MongoDB**** - Database running on port 27017

Initial architecture diagram: [View

Diagram](<https://drive.google.com/file/d/1d3Y6QBUCd5a3VxOU8m44NdSBQ9p6MLWa/view?usp=sharing>)

Prerequisites

- Docker Desktop installed
- Git installed
- Hugging Face account with API token (free)

Setup & Running the Application

1. Clone the repository

```
``bash
```

```
git clone https://github.com/Elocin-MMDC/Brainbytes-Multi-Container.git
```

```
cd Brainbytes-Multi-Container
```



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...

2. Set up your Hugging Face token

Create a `.env` file in the project root:

...

```
HUGGINGFACE_TOKEN=your_hugging_face_token_here
```

...

To get a free token:

1. Sign up at <https://huggingface.co>
2. Click your avatar → Access Tokens
3. Create a fine-grained token with ****Inference**** permissions:
 - Make calls to inference providers
 - Make calls to Inference Endpoints
4. Copy the token into your `.env` file

3. Build and start the containers

```
```bash
```

```
docker-compose up --build
```



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...

### ### 4. Open your browser

- Chat Interface: <http://localhost:8080>
- Profile Page: <http://localhost:8080/profile>
- Dashboard: <http://localhost:8080/dashboard>
- API: <http://localhost:3000>

### ### 5. To stop the application

```
```bash
```

```
docker-compose down
```

...

How to Use

Getting Started

1. Go to **[**http://localhost:8080/profile**](http://localhost:8080/profile)**
2. Create a profile with your name, email, and preferred subjects
3. Click **[***"Use this profile"***](#)** to set it as active
4. Go to **[**http://localhost:8080**](http://localhost:8080)** to start chatting



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5. Your name and preferred subjects will appear in the chat!

Testing the AI Features

****Question Type Detection (same topic, different responses):****

...

What is photosynthesis? → definition response

Explain photosynthesis → explanation response

Give me an example of photosynthesis → example response

...

****Sentiment Analysis:****

...

I'm so frustrated! → empathetic response

I'm confused about gravity → clearer explanation

Thank you, this is great! → positive acknowledgment

...

****Math Solver:****

...



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What is $5 + 3$? → The answer is 8

What is $1+1$? → The answer to $1+1$ is 2

...

****Other topics:****

...

What is algebra?

Who is Jose Rizal?

What is an atom?

Define noun

Explain verb

What is the capital of the Philippines?

...

API Documentation

AI Chat

| Method | Endpoint | Description |

|-----|-----|-----|

| POST | `/api/ai/chat` | Send message; returns AI response with subject, question type, and sentiment |



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****Request:****

```json

```
{ "message": "What is algebra?", "subject": "Math" }
```

```

****Response:****

```json

```
{
```

```
 "userMessage": "What is algebra?",
```

```
 "aiResponse": "Algebra is a branch of mathematics...",
```

```
 "subject": "Math",
```

```
 "questionType": "definition",
```

```
 "sentiment": "neutral"
```

```
}
```

```

User Profiles (CRUD)

| Method | Endpoint | Description |

|-----|-----|-----|



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| POST | `/api/users` | Create a new user profile |

| GET | `/api/users` | Get all user profiles |

| GET | `/api/users/:id` | Get a single user by ID |

| PUT | `/api/users/:id` | Update a user profile |

| DELETE | `/api/users/:id` | Delete a user profile |

****Request body:****

```json

{

"name": "Sarah Nicole Hular",

"email": "sarah@example.com",

"preferredSubjects": ["Math", "Science"]

}

```

Learning Materials

| Method | Endpoint | Description |

|-----|-----|-----|

| POST | `/api/materials` | Create a new learning material |

| GET | `/api/materials` | Get all materials (optional `?subject=Math`) |



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| GET | `/api/materials/:id` | Get a single material by ID |

****Request body:****

```json

{

  "subject": "Math",

  "topic": "Quadratic Equations",

  "content": "A quadratic equation is..."

}

```

Messages & Dashboard

| Method | Endpoint | Description |

|-----|-----|-----|

| GET | `/api/messages` | Get chat history (optional `?subject=Math`) |

| GET | `/api/dashboard/recent` | Get recent activity, total messages, and subject counts |

Database Schema Design

Message Collection



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Stores all chat conversations with AI metadata.

```
```javascript
{
 text: String, // user's question

 response: String, // AI's answer

 subject: String, // detected/selected subject (Math, Science, History, English, Filipino,
General)

 questionType: String, // definition | explanation | example | general

 sentiment: String, // frustrated | confused | positive | neutral

 createdAt: Date

}
```
```

UserProfile Collection

Stores user account information and learning preferences.

```
```javascript
{
 name: String, // required

 email: String, // required, unique

 preferredSubjects: [String], // array of preferred subjects

 createdAt: Date,
```



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updatedAt: Date

}

...

### ### LearningMaterial Collection

Stores learning content organized by subject and topic.

```javascript

{

subject: String, // required (Math, Science, History, English, Filipino)

topic: String, // required (specific topic name)

content: String, // required (the learning content)

createdAt: Date

}

...

AI Implementation

Hybrid Approach (Hugging Face API + Intelligent Fallback)

The AI uses a ****hybrid approach****:



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1. **Primary**: Calls Hugging Face Inference API (`facebook/bart-large-cnn`) for AI-generated responses
2. **Fallback**: Uses local intelligent responses when API is unavailable or rate-limited

The AI logic is in `services/aiService.js` which exports:

- `initializeAI()` - validates environment and token
- `generateResponse(question)` - main hybrid logic
- `detectQuestionType(question)` - detects definition/explanation/example/general
- `detectSentiment(question)` - detects frustrated/confused/positive/neutral

Note on Node.js Compatibility

The backend uses Node.js 14. `AbortController` is not natively available in Node.js 14, so the API timeout handling was adjusted for compatibility. The fallback system ensures the app always responds.

AI Features

1. Expanded Knowledge Base

Each topic has three response variants (definition, explanation, example):

- **Math**: algebra, geometry
- **Science**: photosynthesis, gravity, atoms, evaporation, precipitation
- **History**: Jose Rizal, Philippines



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- **English**: nouns, verbs

2. Question Type Detection

- **Definition**: triggered by "what is", "define", "meaning of"
- **Explanation**: triggered by "how", "why", "explain", "describe"
- **Example**: triggered by "example", "show me", "give me a sample"
- **General**: default for other questions

3. Sentiment Analysis

- **Frustrated** ("frustrated", "frustrating", "hate", "stupid", "ugh") → empathetic prefix
- **Confused** ("confused", "don't understand", "lost", "stuck") → clearer explanation prefix
- **Positive** ("thanks", "great", "love", "helpful") → positive acknowledgment
- **Neutral** → default response style

4. Subject Auto-Classification

Auto-classifies messages into Math, Science, History, English, Filipino, or General for filtering and analytics.

5. Math Expression Solver

Evaluates expressions like "What is 5 + 3?" and returns direct numeric answers.



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Frontend Features

Chat Page (/)

- AI conversation with typing indicator
- ****Subject filter dropdown**** (All, Math, Science, History, English, Filipino, General)
- ****Quick subject buttons**** based on active user's preferred subjects
- ****Active user banner**** showing name and preferred subjects
- Each AI response shows Subject, Question Type, and Sentiment labels
- Personalized welcome message and username in chat

Profile Page (/profile)

- Create profiles with name, email, and preferred subjects
- ****"Use this profile" button** to set active user
- Active profile highlighted with green border and checkmark
- Edit and delete profiles
- Logout button to switch users
- Active profile preferences sync to Chat page

Dashboard Page (/dashboard)

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- Total conversation count
- Per-subject conversation breakdown
- Recent activity feed (last 10 conversations)
- Question Type and Sentiment per conversation

Project Structure

...

brainbytes-multi-container/

```
|— backend/
| |— config/
| |  |— db.js          # MongoDB connection
| |— controllers/
| |  |— materialController.js
| |  |— messageController.js
| |  |— userController.js
| |— models/
| |  |— LearningMaterial.js
| |  |— Message.js
| |  |— UserProfile.js
```



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```
| └─ routes/
|   | └─ materialRoutes.js
|   | └─ messageRoutes.js
|   | └─ userRoutes.js
|   └─ services/
|     └─ aiService.js    # Hugging Face + fallback AI
|   └─ Dockerfile
|   └─ package.json
|     └─ server.js      # Entry point
└─ frontend/
    | └─ pages/
    |   | └─ index.js    # Chat page
    |   | └─ profile.js  # Profile management
    |   └─ dashboard.js  # Activity dashboard
    |   └─ Dockerfile
    |   └─ package.json
    └─ docker-compose.yml
    └─ .env              # Hugging Face token (gitignored)
    └─ .gitignore
    └─ README.md
```



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...

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Development Environment Setup Verification

Development Environment Setup Verification

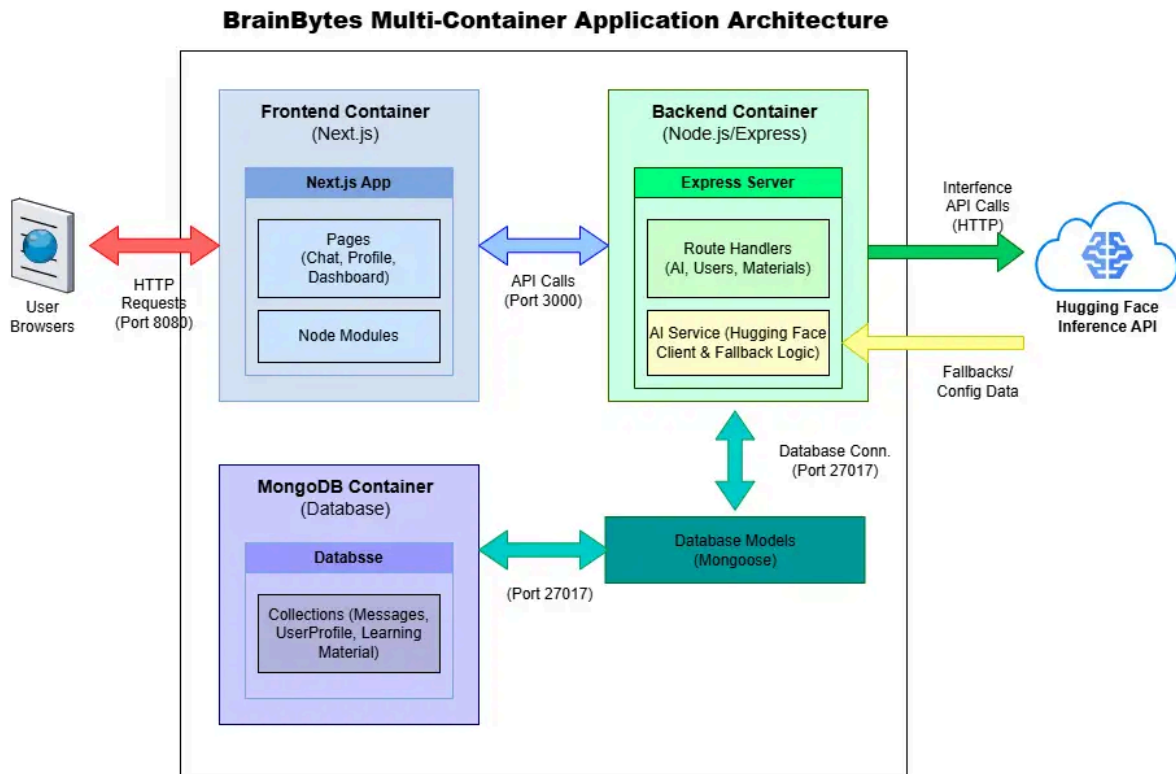
| Team Member | Docker Installed | Git installed | VS Code Installed | Can Run Hello World Container |
|------------------------------|------------------|---------------|-------------------|-------------------------------|
| Sarah Nicole Hular | ✓ | ✓ | ✓ | ✓ |
| Mara Julianne Rose Cervantes | ✓ | ✓ | ✓ | ✓ |
| Eldan Eunice Sinsuan | ✓ | ✓ | ✓ | ✓ |
| Christine Joy Cortes | ✓ | ✓ | ✓ | ✓ |
| Michelle Joi Quesada | ✓ | ✓ | ✓ | ✓ |

Docker Version Information

Docker version 29.4.1, build 055a478



Sample Architecture



Task Distribution Plan

Milestone 1 Task Distribution

Week 1: Container Basics

- **Eldan Eunice Sinsuan:** Set up project repository and basic documentation
- **Michelle Joi Quesada:** Research and document containerization approach
- **All Team Members:** Complete Docker installation and verification

Week 2: Platform Development

- **Sarah Nicole Hular:** Implement frontend container (Next.js)
- **Christine Joy Cortes:** Implement backend container (Node.js)
- **Mara Julienne Rose Cervantes:** Configure MongoDB Atlas and connection

Week 3: Platform Development (continued)

- **Sarah Nicole Hular:** Implement chat interface frontend
- **Christine Joy Cortes:** Implement backend API endpoints
- **Michelle Joi Quesada:** Set up container networking

Week 4: Integration and Testing

- **Eldan Eunice Sinsuan:** Integrate AI model
- **Sarah Nicole Hular:** Implement backend database operations and message history storage
- **Mara Julienne Rose Cervantes:** Create project documentation
- **All Team Members:** Final testing and preparation for submission